



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2024
3rd Year 2nd Semester, Session: 2021-2022

Course Code: CSE-3201
Course Title: Artificial Intelligence and Machine Learning

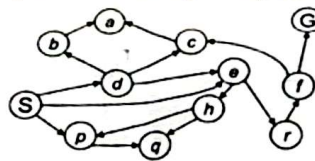
Time: 3.00 hours
Marks: 72

[N.B. Answer any 6 set questions from the following 8 set questions. The figures in the margin indicate full marks.]

1. a) Define Artificial Intelligence (AI) and explain its main types with examples. 2+3
b) Explain problem formulation in AI. Describe the components of a problem definition using a real-life example. 3
c) When performing an action, how a reflex agent and a planning agent will behave? Explain with example. 4

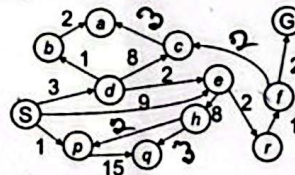
2. a) What is a search problem in AI and what are the main components of a search problem? 3
b) Explain state space graph and search tree in a search problem. 3
c) Consider a Tic-Toe game with 6
• Agent positions: 120
• Food count: 30
• Ghost positions: 12
• Agent facing: NSEW
Now, calculate the followings
• Number of World states
• Number of states for pathing
• Number of states for eat-all-dots

3. a) Apply Depth-First Search (DFS) on the following example. 7



- b) Answer the following questions for Breadth-First Search (BFS) 3
(i) How much space does the fringe take?
(ii) Is it complete?
(iii) Is it optimal?
c) What is alpha beta pruning? How does it improve the efficiency of minimax search? 2

4. a) What is the difference between uninformed search and informed search? 3
b) Apply UCS on the following example. 7



- c) Explain the concept of uncertainty in AI. 2

5. a) Draw a biologically inspired artificial neuron and explain its mathematical model. 3
b) Draw a neural network with proper weight values to perform a logical OR operation. 3
c) Can we perform a logical XOR operation with the neural network mentioned in question 5(a)? 6
If not, what is the reason behind it, and what modifications are needed to solve that problem? Explain with an example.

6. a) Explain the working of the k-Nearest Neighbors (k-NN) algorithm. What are its advantages and limitations? 5
- b) What is overfitting in machine learning? How can it be prevented? 3
- c) What are genetic algorithms? Explain their working with suitable examples. 4

7. a) You are given the following data samples for the above-mentioned problem. Consider a linear model for it. What will be the training losses when $W = 0, 1, 2, 3$, and 4 in case of MSE loss? 5

Dosage (milligram)	Blood Pressure (Diastolic)
5	90
13	92
19	93
10	91

- b) What is learning rate in Machine learning? Discuss the problems associated with learning rate and how can we choose a better learning rate? 4
- c) Describe the process of learning decision trees. What are the advantages and limitations of decision trees? 3

8. Suppose, we have a dataset that has two features for each data sample.

- a) Design a linear model to solve this problem. 2
- b) Draw the computational graph. 2
- c) Assume that the two learnable parameters are W_1 and W_2 , and the loss function is MSE. Find $\frac{\partial Loss}{\partial W_1}$ and $\frac{\partial Loss}{\partial W_2}$ when $X_1=2$, $X_2=3$, $Y=7$. Initialize W_1 and W_2 with 3 and 2, respectively. 8



Department of Computer Science and Engineering

Jashore University of Science and Technology
3rd Year 2nd Semester Final Examination-2025
Session: 2021-22

Course Code: CSE 3205
Course Title: Peripherals, Interfacing and IoT

Time: 3.00 hours
Marks: 72

(Note: Answer any six set of questions from the following eight set of questions)

1. a) What are peripheral devices, and how are the peripheral devices connected to a typical computer system? Explain with a proper diagram. 4
b) Define a peripheral adapter. Explain how a peripheral adapter connects to the data highway with appropriate diagram. 4
c) Describe the different types of system buses and their functions. 4
2. a) Based on the number of bytes transferred, data transfer between a computer and peripheral devices can be classified into two types. Explain these types. 2
b) What is programmed I/O? Explain how a character is transmitted using the Intel 8251 USART interface. Describe the role of the status register and explain how the transmitter readiness is checked before sending data. 1+4
c) Explain the concept of software polling. Write an assembly language program to implement software polling for three peripheral devices using programmed I/O. Assume that STAT1, STAT2, and STAT3 are the addresses of the status registers of the three devices. If a device is ready, call the corresponding input routines PROC1, PROC2, and PROC3. If no device is ready, continue polling. 1+4
3. a) What is Interrupt I/O? How does it improve effectiveness of the processor? 2
b) What is a nested interrupt? Explain how nested interrupts allow a higher-priority device to be serviced while a lower-priority device is already being serviced. 3
c) Define Interrupt Vector Table (IVT). Explain the working principle of the daisy chain method used for handling priority interrupts with a block diagram. 4
d) What is memory-mapped I/O? Describe the advantages and disadvantages of memory mapped I/O. 3
4. a) What is Bluetooth Low Energy (BLE)? How does it conserve power compared to classic Bluetooth? 3
b) Explain BLE advertising and connection process with a state diagram. 4
c) Compare RFID vs NFC in terms of frequency, range, data rate, and applications. 5
5. a) What is a serial interface? 2
b) Write short notes on the following data conversion methods: 4
i. Serial-parallel conversion ii. Parallel-serial conversion
c) Define bit stuffing. Suppose device X wants to send a data frame to device Y using High-Level Data Link Control (HDLC) format. Considering the following values and calculate the final HDLC frame to be sent. 6
Address= 11111100
Control= 00110011
Data= 11111000 11111100
Frame Check Sequence= 11001100 10101010

6. a) Explain the working principle of a Daisy Wheel Printer. 4
b) A 7-segment display is widely used in digital clocks. Explain how a 7-segment display represents numerical digits and mention two practical applications of this type of display. 4
c) Compare Electrostatic Printers and Laser Printers based on Printing principle, Image formation method, and Speed or quality. 4
7. a) Define the Internet of Things (IoT) and explain the differences between IoT, M2M (Machine-to-Machine), and traditional internet systems. 4
b) A smart irrigation system uses soil moisture sensors to water crops automatically. Draw a block diagram of this IoT system and explain how data flows from sensors to actuators. 4
c) Suppose a smart home IoT system suffers from frequent device failures due to poor connectivity. Analyze the possible causes and suggest two strategies to improve reliability. 4
8. a) Define Fog Computing. How does it complement cloud computing in IoT? 3
b) Explain Data Aggregation techniques in IoT networks to reduce traffic. 4
c) Discuss Security Challenges in IoT and suggest lightweight cryptographic solutions. 5



Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2025
3rd Year 2nd Semester, Session: 2021-22

Course Code: CSE 3209

Course Title: Simulation and Modeling

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

- 1 (a) What is a System? Describe ways to study a system. 6
- (b) Describe the Components and Organization of a Discrete-Event Simulation Model. 6
- 2 (a) How Common Random Numbers (CRNs) can be effective as a Variance Reduction Technique? 4
- (b) Explain Naylor and Finger Simulation Model Validation process. 4
- (c) Differentiate between Deterministic and Stochastic Simulation Models. 4
- 3 (a) What are the basic differences between analytical solution and simulation? 4
- (b) Discuss the components that are found in most discrete-event simulation model. 8
- 4 (a) Describe simulation of a Single-Server Queueing System. 6
- (b) Prove that the variance of Exponential Distribution is β^2 . 6
- 5 (a) In a Single-Server Queueing System, arrivals occur at times 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8, and 7.2. Departures (service completions) occur at times 2.0, 3.0, 4.0, 6.0, and 9.0, and the simulation ends at time $T(n) = 9.0$. Compute $q^*(n)$. 8
- (b) What are the desirable features of a Simulation Software? 4
- 6 (a) Describe simulation of an Inventory System. 6
- (b) Why and how Inverse Transform is used to generate random variates in exponential distribution? 6
- 7 (a) What properties should a "good" arithmetic random-number generator possess? 4
- (b) How can we maximize the period of a Random Number Generator beyond Linear Congruential Generators? 8
- 8 (a) Describe Bernoulli and Discrete Uniform distributions. 4
- (b) What is Model Verification and Validation? Describe? 4
- (c) What statistical technique(s) do we use nonstationary and autocorrelated for output data analysis? 4

----- 0 -----



Jashore University of Science and Technology

Department of Computer Science and Engineering

Bachelor of Engineering

3rd Year 2nd Semester Final Examination-2026 (Session: 2021-2022)

Course Code: AIS 3201

Course Title: Accounting Information System

Total Time: 3:00 Hours

Total Marks: 72

Instructions and Information:

- Answer any six (06) sets from the following set of questions.
- Answer all parts (a, b, c and so on) of each question consecutively.
- Start answering each question and sub-question from a fresh page.
- Answer should be clearly numbered with the sub-question, and then ruled off.
- The numbers in the right margin indicate assigned marks for each part of a question.

1. a) What is an accounting system? 2
- b) "Book-keeping and Accounting are the same thing." Do you agree? Explain. Write the differences between Book-keeping and Accounting. 5
- c) How do accurate financial practices contribute to transparency, investor confidence, and the prevention of corporate fraud? 5
2. a) List the components of Owner's Equity. 2
- b) Bipra started her own consulting firm, Alia Enterprise, on March 1, 2024. The following transactions occurred during the month of March. 10

- March 1 Bipra invested Tk. 15,000 cash in the business.
- 2 Paid Tk. 1,200 for office rent for the month.
- 3 Purchased Tk. 1600 of supplies on credit.
- 5 Paid Tk. 1,250 to advertise in the Daily Star.
- 8 Received Tk. 4,500 cash for services performed.
- 12 Withdrew Tk. 2,000 cash for personal use.
- 15 Performed Tk. 5,400 of services on account.
- 17 Paid Tk. 2,500 for employees' salaries.
- 20 Paid for the supplies purchased on account on March 3.
- 22 Received a cash payment of Tk. 4,000 for services performed on account on March 15.
- 26 Borrowed Tk. 5,000 from the bank on a note payable.
- 29 Purchased furniture for Tk. 6,000 on account.
- 30 Paid Tk. 500 for utilities.

Instructions:

Prepare the tabular analysis from the above transactions on the accounting equation.

3. a) "Ledger is the king of all accounts."- Explain the statement. 4
- b) Imrul Kayees is a licensed dentist. During the first month of the operation of his business, the following events and transactions occurred. 8

April 1	Invested Tk. 20,000 cash in his business.
3	Hired a secretary receptionist at a salary of Tk. 800 per week payable monthly.
6	Paid office rent in advance for the month Tk. 1,100.
10	Purchased dental supplies on credit from Smile Company Tk. 4,000.
15	Performed dental services Tk. 10,000 in which of Tk. 6,000 are received in cash.
20	Paid Smile Company for accounts payable due.
30	Paid secretary receptionist for the month Tk. 3,200.

Instructions:

- Journalize the transactions
- Post to the ledger accounts.

4. a) Discuss the differences between financial and managerial accounting. 4

- b) "The variable cost per unit varies with output, whereas the fixed cost per unit does not." Do you agree? Explain why or why not. 4
- c) Define the following terms with example: conversion cost, opportunity cost, and sunk cost. 4

5. a) Define margin of safety in CVP analysis. 2
- b) Voltar Company manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below: 10

	Total	Per Unit	Percent of Sales
Sales (20,000 units)	Tk.1,200,000	Tk.60	100%
Variable expenses	900,000	45	? %
Contribution margin	300,000	Tk.15	? %
Fixed expenses	240,000		
Net operating income	Tk. 60,000		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

Required:

- Compute the company's CM ratio and variable expense ratio.
- Compute the company's break-even point in both units and sales dollars. Use the equation method.
- Assume that sales increase by Tk. 400,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase? Use the CM ratio to compute your answer.
- Refer to the original data. Assume that next year management wants the company to earn a profit of at least Tk. 90,000. How many units will have to be sold to meet this target profit?
- Refer to the original data. Compute the company's margin of safety in both dollar and percentage form.

6. a) Define manufacturing and non-manufacturing costs with examples. 3
- b) X company has gathered the following information. 9

Units in beginning work in process	0
Units started into production	TK. 40,000
Units in ending work in process	TK. 6,000
Percent complete in ending work in process:	
Conversion costs	40%
Materials	100%
Costs incurred:	
Direct materials	TK. 72,000
Direct labour	TK. 81,000
Overhead	TK. 1,01,000

Instructions:

- Compute equivalent units of production for materials and for conversion costs
- Determine the unit costs of production.
- Show the assignment of costs to units transferred out and in process.

7. a) Define Accounting Information System (AIS). 3
- b) What are the major components of an Accounting Information System (AIS), and how do they work together to improve the quality of financial information? 5
- c) What are the key differences and interrelationships between Accounting Information Systems (AIS) and Management Information Systems (MIS) in terms of their objectives, components, and contributions to organizational decision-making? 4

8. a) Explain why organizations are shifting from traditional file systems to Database Management Systems (DBMS). 4
- b) What are Data Definition Language (DDL) and Data Manipulation Language (DML) in database systems? 4
- c) Analyze how AIS, MIS, and DBMS collectively contribute to efficient business processes and decision-making. 4

- b) "The variable cost per unit varies with output, whereas the fixed cost per unit is constant." Do you agree? Explain why or why not. 4
- c) Define the following terms with example: conversion cost, opportunity cost, and sunk cost. 4
5. a) Define margin of safety in CVP analysis. 2
- b) Voltar Company manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below: 10

	Total	Per Unit	Percent of Sales
Sales (20,000 units)	Tk.1,200,000	Tk.60	100%
Variable expenses	900,000	45	? %
Contribution margin	300,000	Tk.15	? %
Fixed expenses	240,000		
Net operating income	Tk. 60,000		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

Required:

- Compute the company's CM ratio and variable expense ratio.
- Compute the company's break-even point in both units and sales dollars. Use the equation method.
- Assume that sales increase by Tk. 400,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase? Use the CM ratio to compute your answer.
- Refer to the original data. Assume that next year management wants the company to earn a profit of at least Tk. 90,000. How many units will have to be sold to meet this target profit?
- Refer to the original data. Compute the company's margin of safety in both dollar and percentage form.

6. a) Define manufacturing and non-manufacturing costs with examples. 3
- b) X company has gathered the following information. 9

Units in beginning work in process	0
Units started into production	TK. 40,000
Units in ending work in process	TK. 6,000
Percent complete in ending work in process:	
Conversion costs	40%
Materials	100%
Costs incurred:	
Direct materials	TK. 72,000
Direct labour	TK. 81,000
Overhead	TK. 1,01,000

Instructions:

- Compute equivalent units of production for materials and for conversion costs
- Determine the unit costs of production.
- Show the assignment of costs to units transferred out and in process.

7. a) Define Accounting Information System (AIS). 3
- b) What are the major components of an Accounting Information System (AIS), and how do they work together to improve the quality of financial information? 5
- c) What are the key differences and interrelationships between Accounting Information Systems (AIS) and Management Information Systems (MIS) in terms of their objectives, components, and contributions to organizational decision-making? 4
8. a) Explain why organizations are shifting from traditional file systems to Database Management Systems (DBMS). 4
- b) What are Data Definition Language (DDL) and Data Manipulation Language (DML) in database systems? 4
- c) Analyze how AIS, MIS, and DBMS collectively contribute to efficient business processes and decision-making. 4